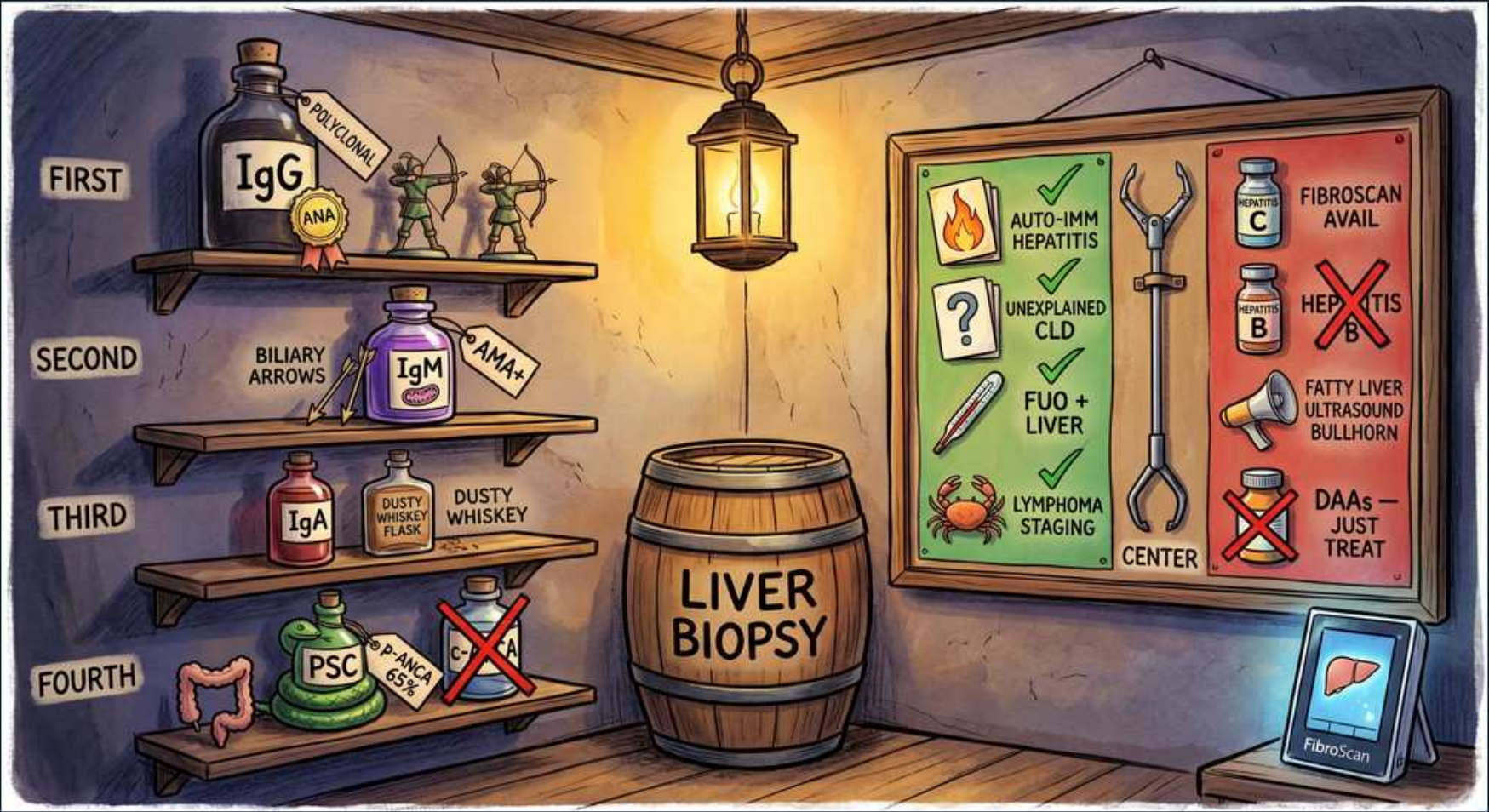


Liver Approach — Cellar Shelves



1. FIRST shelf — IgG

WHAT YOU SEE

Top 'FIRST' shelf, big IgG bottle, 'POLYCLONAL', ANA ribbon

WHAT IT ENCODES

Autoimmune hepatitis type 1 (AIH-1) is a T-cell-mediated interface hepatitis with a polyclonal hypergammaglobulinemia: total IgG is elevated (often >1.5x ULN) and ANA is the classic serologic marker. AIH-1 is the adult form (bimodal: peri-puberty and ~40s), strongly female-predominant, and HLA-DR3/DR4 associated. Histology shows interface hepatitis with a plasma-cell-rich lymphocytic infiltrate; rosetting of hepatocytes is characteristic. First-line treatment is corticosteroids (prednisone ~30-40 mg/day taper) plus azathioprine as a steroid-sparing agent; budesonide is an alternative in non-cirrhotic patients.

BOARD TRAP

WRONG answer: ordering anti-LKM1 first or starting ursodeoxycholic acid. UDCA treats PBC, not AIH. Discriminator: high IgG + ANA/ASMA + interface hepatitis on biopsy points to AIH-1, which needs immunosuppression (steroids + azathioprine), NOT UDCA. Anti-LKM1 defines type 2 AIH (children), not the classic adult ANA-positive type 1.

2. ASMA archers

WHAT YOU SEE

Two archers (anti-smooth-muscle) on first shelf

WHAT IT ENCODES

Anti-smooth-muscle antibody (ASMA), whose target antigen is filamentous (F-)actin, is the second key first-line marker of AIH type 1; anti-actin titers $\geq 1:80$ are most specific. ASMA and ANA together (often with high IgG) define AIH-1 and are part of the simplified IAIHG diagnostic score (ANA or ASMA $\geq 1:40$ scores 1 point, $\geq 1:80$ scores 2). The combination markedly raises diagnostic probability and supports treatment with steroids plus azathioprine even before biopsy when the score is high.

BOARD TRAP

WRONG answer: calling ASMA specific for PBC or PSC. Discriminator: ASMA (anti-F-actin) is an AIH-1 marker. AMA is the PBC marker and p-ANCA the PSC marker; do not assign ASMA to the cholestatic diseases.

3. Biliary arrows

WHAT YOU SEE

'SECOND' shelf, 'BILIARY ARROWS'

WHAT IT ENCODES

The biliary arrows mark the cholestatic pattern: an alkaline phosphatase (ALP)/GGT-predominant rise (rather than ALT/AST-predominant) directs you toward bile-duct disease, classically PBC on this second shelf. A confirmed hepatic source of ALP is supported by a concurrently elevated GGT. In a middle-aged woman with cholestatic LFTs and pruritus/fatigue, the next step is testing AMA.

BOARD TRAP

WRONG answer: attributing an isolated ALP elevation to hepatic disease without checking GGT. Discriminator: ALP also rises from bone, placenta, and gut. A normal GGT with high ALP points to a non-hepatic (e.g., bone) source, not PBC. Fractionate or check GGT before pursuing a biliary workup.

4. IgM jar (PBC)

WHAT YOU SEE

Purple IgM jar on second shelf

WHAT IT ENCODES

PBC characteristically elevates IgM (in contrast to the IgG elevation of AIH), reflecting its autoimmune destruction of small intralobular bile ducts (florid duct lesion with granulomas on biopsy). PBC affects middle-aged women (~90% female, 40-60s) presenting with fatigue and pruritus; it associates with other autoimmune disease, especially Sjogren and Hashimoto thyroiditis. Mnemonic: PBC = IgM and Middle-aged; AIH = IgG.

BOARD TRAP

WRONG answer: expecting elevated IgG in PBC or elevated IgM in AIH. Discriminator: PBC = IgM, AIH = IgG. Swapping the immunoglobulin class is a classic distractor; pair IgM with AMA-positive cholestatic disease in a middle-aged woman.

5. AMA+ tag

WHAT YOU SEE

'AMA+' tag on second shelf

WHAT IT ENCODES

Anti-mitochondrial antibody (AMA), targeting the E2 subunit of pyruvate dehydrogenase (PDC-E2), is the hallmark of primary biliary cholangitis, present in ~90-95% with high specificity (>95%). A cholestatic ALP elevation plus AMA-positivity is enough to diagnose PBC without biopsy. First-line therapy is ursodeoxycholic acid (UDCA) 13-15 mg/kg/day, which improves survival and slows progression; obeticholic acid or fibrates are second-line for inadequate UDCA responders.

BOARD TRAP

WRONG answer: doing a liver biopsy to confirm AMA-positive PBC, or starting steroids. Discriminator: in a patient with cholestatic LFTs and positive AMA, biopsy is unnecessary; treat with UDCA. Steroids/azathioprine are for AIH, not PBC.

6. IgA jar (third)

WHAT YOU SEE

'THIRD' shelf, IgA jar

WHAT IT ENCODES

Elevated IgA is the immunoglobulin clue for alcoholic (and to a degree, non-alcoholic fatty) liver disease. Alcoholic hepatitis classically shows AST:ALT >2:1 (often with AST and ALT both <300-400 IU/L), elevated GGT, and macrocytosis. Mnemonic by immunoglobulin: IgG = AIH, IgM = PBC, IgA = Alcohol.

BOARD TRAP

WRONG answer: interpreting an AST:ALT ratio >2 as viral or autoimmune hepatitis. Discriminator: AST:ALT >2:1 with high GGT and elevated IgA suggests alcohol; viral/AIH typically have ALT > AST. Markedly elevated transaminases (>1000) instead suggest ischemic, toxic (acetaminophen), or acute viral injury, not alcohol.

7. Dusty whiskey flask

WHAT YOU SEE

'DUSTY WHISKEY FLASK / DUSTY WHISKEY'

WHAT IT ENCODES

The dusty whiskey flask anchors alcohol-related liver disease, the clinical partner of the IgA elevation. Severity in alcoholic hepatitis is graded by the Maddrey discriminant function (DF ≥ 32) or MELD; DF ≥ 32 indicates severe disease that may benefit from prednisolone 40 mg/day for 28 days, with the Lille score at day 7 assessing response (Lille >0.45 = non-responder, stop steroids). Pentoxifylline is no longer favored.

BOARD TRAP

WRONG answer: giving steroids to every alcoholic hepatitis patient. Discriminator: corticosteroids are reserved for SEVERE disease (Maddrey DF ≥ 32) and are contraindicated with active GI bleeding, uncontrolled infection/sepsis, or AKI. Mild disease (DF <32) is managed with abstinence and nutrition, not steroids.

8. PSC bowel (fourth)

WHAT YOU SEE

'FOURTH' shelf, bowel/colon model, PSC

WHAT IT ENCODES

Primary sclerosing cholangitis (PSC) is a fibro-obliterative inflammation of intra- and extrahepatic bile ducts, strongly linked to inflammatory bowel disease (~70-80%, predominantly ulcerative colitis). Unlike PBC, PSC is male-predominant (~2:1) and presents in younger men (~30-40s) with cholestatic LFTs. Diagnosis is by MRCP showing multifocal strictures with intervening dilation — the classic 'beads-on-a-string' appearance; no medical therapy improves survival, and UDCA is not recommended for survival benefit.

BOARD TRAP

WRONG answer: starting UDCA to improve survival in PSC, or assuming a normal colonoscopy excludes IBD. Discriminator: UDCA has no proven survival benefit in PSC (high-dose UDCA may be harmful). Also, every new PSC patient needs a colonoscopy because IBD is often silent, and PSC-IBD carries a high colorectal cancer risk requiring annual surveillance.

9. p-ANCA 85%

WHAT YOU SEE

'p-ANCA 85%' tag

WHAT IT ENCODES

Atypical perinuclear ANCA (p-ANCA, more precisely 'atypical p-ANCA'/peripheral anti-nuclear neutrophil antibody) is positive in ~80% of PSC but is nonspecific (also seen in AIH and UC), so it does not make the diagnosis. PSC is a diagnosis of MRCP (or ERCP) showing beaded multifocal strictures plus a cholestatic ALP elevation. PSC patients face a markedly increased lifetime risk of cholangiocarcinoma (~10-15%, screen with MRCP + CA 19-9 yearly) and gallbladder cancer.

BOARD TRAP

WRONG answer: diagnosing PSC on a positive p-ANCA, or doing a biopsy to confirm. Discriminator: p-ANCA is nonspecific; the diagnosis is made by cholangiographic imaging (MRCP first-line). Biopsy showing 'onion-skin' periductal fibrosis is supportive but not required and is often nondiagnostic due to patchy disease.

10. AMA crossed (PSC)

WHAT YOU SEE

AMA vial with red X near PSC

WHAT IT ENCODES

PSC is characteristically AMA-NEGATIVE, the key serologic discriminator from PBC. The cleanest separation: PBC = AMA-positive, female, IgM, small intrahepatic ducts, treat with UDCA; PSC = AMA-negative, male, p-ANCA, IBD-associated, large-duct beading on MRCP, no effective medical therapy. ERCP with dilation/stenting is reserved for dominant strictures in PSC, and rising ALP or a dominant stricture should prompt cholangiocarcinoma evaluation.

BOARD TRAP

WRONG answer: a positive AMA points to PSC. Discriminator: a positive AMA argues for PBC, not PSC. If a patient with IBD has cholestasis and positive AMA, reconsider PBC (or an overlap); classic PSC is AMA-negative with beaded ducts on cholangiography.

11. Liver biopsy barrel

WHAT YOU SEE

Large central barrel labeled 'LIVER BIOPSY'

WHAT IT ENCODES

Liver biopsy remains the central confirmatory/staging tool when noninvasive workup is inconclusive or when histology changes management (grading inflammation, staging fibrosis, identifying overlap syndromes). Percutaneous biopsy carries small risks (pain, ~1/1000 significant bleeding); a transjugular approach is preferred with coagulopathy, ascites, or when hepatic venous pressure gradient measurement is needed. It is most useful in suspected AIH and in unexplained chronic liver disease.

BOARD TRAP

WRONG answer: defaulting to biopsy when a noninvasive diagnosis is already established (positive AMA + cholestasis = PBC; clear NAFLD on imaging; known viral hepatitis stageable by elastography). Discriminator: biopsy is for diagnostic uncertainty or when results will change management, not routine confirmation of an already-clear diagnosis.

12. Biopsy indications (green)

WHAT YOU SEE

Green panel: autoimmune hepatitis, unexplained CLD, FUO+liver, lymphoma staging

WHAT IT ENCODES

Biopsy IS indicated when histology drives management: (1) suspected autoimmune hepatitis (interface hepatitis confirms and grades it before immunosuppression), (2) unexplained chronic liver disease after a negative noninvasive workup, (3) fever of unknown origin with suspected hepatic involvement (granulomas, infiltrative disease), and (4) staging of suspected hepatic lymphoma or infiltrative malignancy. In these settings tissue answers a question that serology and imaging cannot.

BOARD TRAP

WRONG answer: treating every elevated-LFT patient with biopsy. Discriminator: the green-panel indications share a feature — diagnostic uncertainty that tissue resolves. If serology (e.g., AMA) or imaging already gives the diagnosis, biopsy moves to the red 'avoid' column.

13. FibroScan available (red)

WHAT YOU SEE

Red panel: 'FIBROSCAN AVAIL' hep C vial

WHAT IT ENCODES

When transient elastography (FibroScan) can stage fibrosis noninvasively, avoid biopsy. In chronic hepatitis C and B, elastography (plus serum panels like FIB-4, APRI, or ELF) reliably stratifies fibrosis (cutoffs roughly: F0-1 <7 kPa, cirrhosis >12.5 kPa, though disease-specific). This spares patients an invasive procedure and is the modern standard for fibrosis staging in chronic viral hepatitis and NAFLD.

BOARD TRAP

WRONG answer: biopsying chronic hepatitis C just to stage fibrosis. Discriminator: noninvasive elastography (FibroScan) is preferred for fibrosis staging in viral hepatitis. Biopsy is reserved for discordant noninvasive results or suspected concurrent disease. Note elastography is unreliable with acute hepatitis, congestion, cholestasis, or high BMI/ascites (use XL probe).

14. Hep B / fatty US

WHAT YOU SEE

'HEP B', 'FATTY LIVER ULTRASOUND' bullhorn, crossed

WHAT IT ENCODES

Do not biopsy when the diagnosis is already clear noninvasively: known chronic HBV (diagnosed serologically, staged by elastography) and hepatic steatosis evident on ultrasound (increased echogenicity/'bright liver'). For NAFLD, biopsy is reserved for cases where NASH with significant fibrosis is suspected and would change management, or to exclude competing diagnoses — not for routine confirmation of fatty liver.

BOARD TRAP

WRONG answer: routine biopsy for straightforward NAFLD or known viral hepatitis. Discriminator: routine NAFLD or uncomplicated viral hepatitis does NOT need biopsy. Biopsy in NAFLD is justified only when there is diagnostic uncertainty or suspicion of advanced NASH fibrosis (high FIB-4/elastography) that alters management.

15. DAAs — just treat

WHAT YOU SEE

'DAAs — JUST TREAT' with crossed-out biopsy

WHAT IT ENCODES

In hepatitis C, direct-acting antivirals (DAAs) such as sofosbuvir/velpatasvir or glecaprevir/pibrentasvir achieve SVR (cure) in >95% over 8-12 weeks regardless of fibrosis stage, so you 'just treat' rather than biopsy first. Pre-treatment workup focuses on genotype (often pangenotypic regimens now), cirrhosis status (assessed noninvasively), and prior treatment, not histology. Cirrhotic patients still need HCC surveillance after cure.

BOARD TRAP

WRONG answer: requiring a liver biopsy before treating hepatitis C. Discriminator: DAAs are given irrespective of fibrosis stage, so pre-treatment biopsy is unnecessary. The only fibrosis question that matters is cirrhosis (changes regimen/duration and mandates ongoing HCC surveillance), and that is answered noninvasively.

16. FibroScan tablet

WHAT YOU SEE

Tablet device labeled 'FibroScan'

WHAT IT ENCODES

FibroScan (vibration-controlled transient elastography) measures liver stiffness in kPa as a noninvasive surrogate for fibrosis and is the alternative that spares many patients a biopsy across NAFLD, viral hepatitis, and alcohol-related disease. It also yields a CAP score for steatosis. Reliability requires fasting and is reduced by ascites, acute hepatitis, hepatic congestion, cholestasis, or obesity (use the XL probe).

BOARD TRAP

WRONG answer: trusting a high FibroScan reading in the setting of acute hepatitis or heart failure as proof of fibrosis. Discriminator: inflammation, congestion (right heart failure), cholestasis, and food intake all falsely raise liver stiffness. Confirmatory biopsy is warranted when elastography is unreliable or discordant with clinical/serum markers.